PERSONALIZED PREDICTION OUTPUT AND INTERPRETATION

The developed system generates personalized muscle fatigue predictions based on subject-specific inputs provided by physiotherapists, including age, gender, activity levels, and smoking history (Fig below).

For a sample subject aged 45 years with moderate work activity (level 2), higher spare-time activity (level 3), and no smoking history, the model predicts a Maximum Voluntary Contraction of approximately 609.82.

Using this value, the system computes a fatigue coefficient of 0.00795, indicating the rate at which muscle fatigue develops. Based on this, a safe exercise duration of approximately 87.19 seconds is recommended.

Additionally, the model suggests an optimal contraction level of 69.75 and a recovery (rest) period of 34.88 seconds to minimize fatigue risk. The overall fatigue risk for this subject is classified as moderate. This output demonstrates how the proposed hybrid model can assist physiotherapists in designing personalized and safe exercise plans.

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ENTER SUBJECT DETAILS FOR PERSONALIZED PREDICTION
Age: 45
Gender (M/F): M
Work Activity (1-4): 2
Spare Activity (1-4): 3
Smoking Packyears: 0

----- PERSONALIZED RESULTS -----
Predicted MVC: 609.8191378110594
Fatigue Coefficient: 0.00795
Safe Exercise Time: 87.18832459873525
Recommended Contraction: 69.7506596789882
Recommended Rest: 34.8753298394941
Fatigue Risk: Moderate Risk
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Prediction and Recommendation shown to Physiotherapist based on Patient data

Parameter	Case 1 (Low Risk)	Case 2 (Moderate Risk)	Case 3 (High Risk)
Age (years)	25	45	34
Gender	M	M	F
Work Activity (1–4)	4	2	2
Spare Activity (1–4)	3	3	1
Smoking (Pack-years)	0	0	0
Predicted MVC (N)	560.3	609.8	621.723
Fatigue Coefficient	0.0052	0.00795	0.00718
Safe Exercise Time (secs)	133.04	87.18	96.538
Recommended Contraction (secs)	106.43	69.75	77.2308
Recommended Rest (secs)	53.217	34.87	38.615
Fatigue Risk	Low	Moderate	High